

# Digital Video Colourmaps for Checking the Legibility of Displays by Dichromats

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*Abstract:* We propose replacement colourmaps that allow a designer to check the colours seen by protanopes and deuteranopes. Construction of the colourmaps is based on the LMS specification of the primaries of a standard video monitor and has been carried out for 256 colours, including 216 colours that are common to many graphics applications of MS Windows and Macintosh computing environments.

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*Key words:* colour; dichromacy; computer; simulation; recognition; fundamentals; colour vision deficiencies

## INTRODUCTION

Digital video technologies allow us to create and modify colour images. Most often, the choice of colour is supposed to enhance readability. However, for those who suffer from colour-blindness and who represent 8% of the male population, the choice of colours may not be optimal. Colour-blinds confuse colours that are discriminable for the normal. The most severely affected people are those who have a dichromatic form of colour vision deficiency. It is possible to compute colour confusions and to simulate dichromatic colour vision. Here, we propose colourmaps to replace the “system” ones, and to allow a designer with normal colour vision to simulate the colours seen by dichromats.

## METHODS

The method is based on the LMS system, which specifies colours in terms of the relative excitations of the longwave sensitive (L), the middlewave sensitive (M), and the short-wave sensitive (S) cones. As dichromats lack one class of cone photopigment, they confuse colours that differ only in the excitation of the missing class of photopigment. In contrast to the case of the trichromatic observer, who requires colour specifications by three components, two components are sufficient to specify colour for the dichromat. One can construct a rule to reduce any set of confused colours to a single three-component colour specification.

Quantitative estimates of the colour perceptions typical of protanopic and deuteranopic observers have been given for the whole range of colours in the Munsell colour-order system.<sup>1</sup> Potentialities of computer graphics systems to synthesize a picture of the world as seen by dichromats have been outlined, and dichromatic versions of an image have been produced using  $(u', v')$  chromaticity transformation for every pixel.<sup>2</sup> A more general transformation for simulating colour-deficient vision, which circumvents the CIE XYZ system and comprises both the dichromatic and anomalous case, has been implemented in a colour editor for display design.<sup>3,4</sup>

Here, we construct colourmaps to replace a standard palette of 256 colours, including 216 colours that are common to many graphics applications of MS Windows and Macintosh computing environments, and show how a colour image would look for protanopes or for deuteranopes.

In our former publications,<sup>5,6</sup> we have given an illustration and described a detailed algorithm for simulating co-

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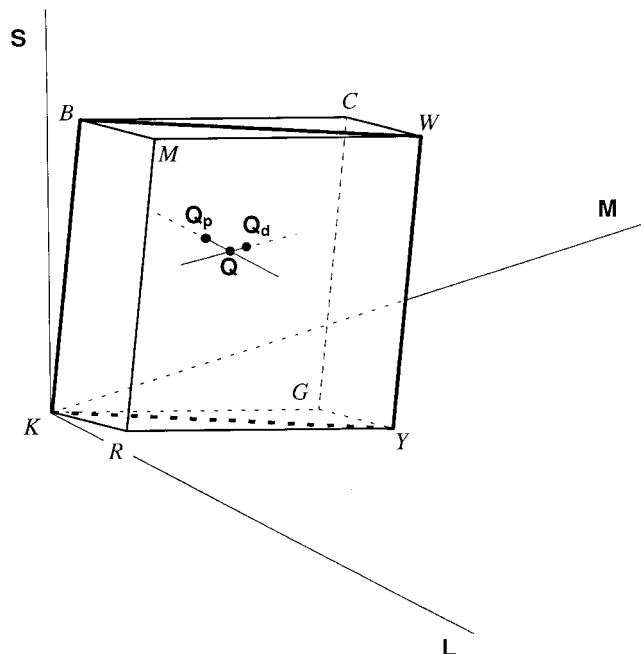


FIG. 1. Representation of stimuli in LMS space. All colours obtainable by combination of the primaries are included in the parallelepiped KBMRGCWY. K: Black. B: Blue primary. M: Magenta. R: Red primary. G: Green primary. C: Cyan. W: White. Y: Yellow. The replacement colours are located in the reduced plane KBWY.  $Q_pQ$  is a confusion line for protanope; all colours of  $Q_pQ$  line reduce to its intersection  $Q$  with the plane KBWY.  $Q_dQ$  is a confusion line for deuteranope; all colours of  $Q_dQ$  line reduce to its intersection  $Q$  with the plane KBWY.

lour appearance for dichromats. We proposed to reduce all stimuli belonging to a confusion line, i.e., to a line parallel to the axis of the missing photopigment in the LMS specification space, to a single colour and to represent the dichromat's colour space by two half-planes, each of which included the neutral axis and an anchor wavelength, which is the wavelength that appears similar to dichromats and

normals. The advantage of specifying colours in the LMS colour space rather than the XYZ colour space is that the transformation takes into account the altered luminosity function of dichromats, especially for protanopes.

Here we maintain this scheme of reduction, but we introduce three compromises in order to achieve the practical goal of replacement colourmaps that can be implemented on any graphics monitor.

First, instead of requiring individual calibration of the video display, we assume that the video display primaries and nominal white are representative of recent standards for Cathode Ray Tube (CRT) monitors<sup>7,8</sup> and that its video-transfer function is a power function with an exponent of 2.2 ("gamma").

Second, the video display standard is specified in terms of CIE 1931 ( $x, y$ ) chromaticity coordinates, but the best sets of fundamentals are not derived from the CIE 1931 colorimetric observer. The Smith and Pokorny<sup>9</sup> set of fundamentals is derived from the Judd-Vos modified colorimetric observer,<sup>10,11</sup> i.e., an observer slightly different from the CIE 1931 one, with no possible linear transformation between them. Because the video display standard does not recommend a spectral power distribution for primaries but only chromaticities, we convert from ( $x, y$ ) chromaticity coordinates to modified ( $x', y'$ ) coordinates using the formula of Vos<sup>11</sup> this formula strictly applies for spectral stimuli, and our second compromise consists in extending it to the primaries and the nominal white.

Third, in order to use as many colours as possible of the video display gamut, we choose replacement colours on a diagonal plane of the RGB colour space of the display device (Fig. 1).

## RESULTS

Starting with a standard palette of 216 colours, which are commonly used with MS Windows and Macintosh computing environments, and which we have extended to 256

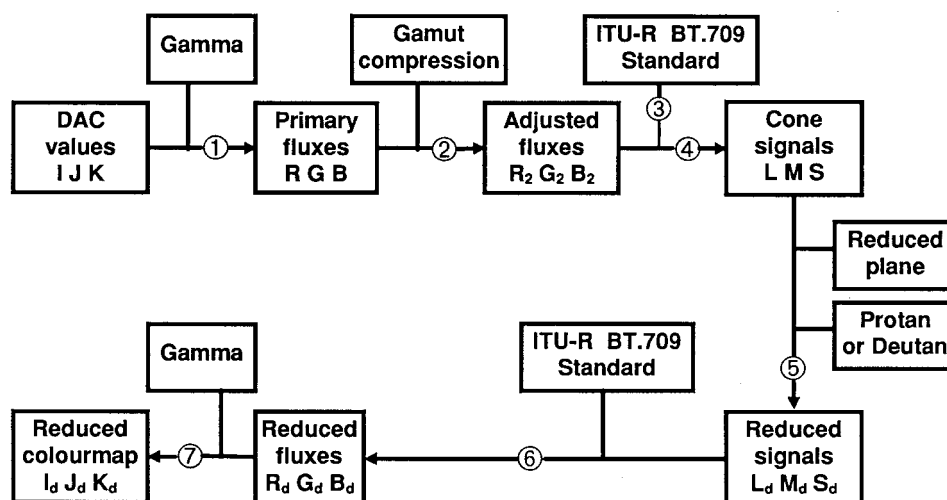


FIG. 2. Computational procedure giving the replacement colourmaps to simulate dichromatic vision (subscript  $d$ ). Each operation is numbered according to the successive steps described in the Methods section.

TABLE I. Chromaticity ( $x, y$ ) of primaries and reference white of the ITU-R BT.709 standard and of the NTSC. Modified chromaticity ( $x', y'$ ) converted from ( $x, y$ ) using Vos (1978) transformation.

|                 | ITU-R BT.709<br>standard |        | NTSC<br>standard |        |
|-----------------|--------------------------|--------|------------------|--------|
|                 | $x$                      | $y$    | $x$              | $y$    |
| Red primary     | 0.64                     | 0.33   | 0.67             | 0.33   |
| Green primary   | 0.30                     | 0.60   | 0.21             | 0.71   |
| Blue primary    | 0.15                     | 0.06   | 0.14             | 0.08   |
| Reference white | D65                      |        | CIE C            |        |
|                 | 0.3127                   | 0.3290 | 0.310            | 0.316  |
|                 |                          |        | D93 white        |        |
|                 |                          |        | 0.2831           | 0.2971 |
|                 | ITU-R BT.709<br>standard |        |                  |        |
|                 | $x'$                     | $y'$   |                  |        |
| Red primary     | 0.6384                   | 0.3326 |                  |        |
| Green primary   | 0.3018                   | 0.6008 |                  |        |
| Blue primary    | 0.1530                   | 0.0682 |                  |        |
| Reference white | D65                      |        |                  |        |
|                 | 0.3157                   | 0.3345 |                  |        |

colours, we have constructed two replacement colourmaps that illustrate protanopic and deuteranopic vision, respectively.

The transformation scheme to compute the replacement colourmaps includes several steps (Fig. 2).

1. Given ( $I, J, K$ ) as the 8-bit DAC values for each of the ( $R, G, B$ ) video channels, we compute the relative photometric quantities  $R, G, B$ :

$$R = (I/255)^{2.2} \quad (1)$$

$$G = (J/255)^{2.2} \quad (1)$$

$$B = (K/255)^{2.2}. \quad (1)$$

2. In order to produce reduced colours that are included in the colour gamut of the monitor, we slightly reduce the colour domain of the initial palette. This is achieved by appropriate scaling of the relative photometric quantities.

For protanopes:

$$R_2 = 0.992052 R + 0.003974 \quad (2)$$

$$G_2 = 0.992052 G + 0.003974 \quad (2)$$

$$B_2 = 0.992052 B + 0.003974. \quad (2)$$

For deuteranopes:

$$R_2 = 0.957237 R + 0.0213814 \quad (2)$$

$$G_2 = 0.957237 G + 0.0213814 \quad (2)$$

$$B_2 = 0.957237 B + 0.0213814. \quad (2)$$

3. The LMS specification of each colour is obtained from the CIE 1931 ( $x, y$ ) specifications of the CRT display by the following procedure. We first apply the Judd-Vos colorimetric modification<sup>10,11</sup>

$$x'_\lambda = \frac{1.0271x_\lambda - 0.00008y_\lambda - 0.00009}{0.03845x_\lambda + 0.01496y_\lambda + 1} \quad (3)$$

$$y'_\lambda = \frac{0.00376x_\lambda + 1.0072y_\lambda + 0.00764}{0.03845x_\lambda + 0.01496y_\lambda + 1} \quad (3)$$

to the chromaticity ( $x, y$ ) of the red, green, and blue primaries and nominal white of the International Telecommunication Union ITU-R BT.709 standard<sup>7</sup> (Table I) and obtain slightly modified chromaticity coordinates ( $x', y'$ ).

Then we compute the corresponding modified ( $X_2, Y_2, Z_2$ ) tristimulus values for the primaries and get the matrix<sup>12</sup>

$$\begin{pmatrix} X_2 \\ Y_2 \\ Z_2 \end{pmatrix} = \begin{pmatrix} RGB\_to\_XYZ \end{pmatrix} \begin{pmatrix} R_2 \\ G_2 \\ B_2 \end{pmatrix} = \begin{pmatrix} 40.9568 & 35.5041 & 17.9167 \\ 21.3389 & 70.6743 & 7.98680 \\ 1.86297 & 11.4620 & 91.2367 \end{pmatrix} \begin{pmatrix} R_2 \\ G_2 \\ B_2 \end{pmatrix}.$$

From Smith and Pokorny,<sup>9</sup> we get

$$\begin{pmatrix} L \\ M \\ S \end{pmatrix} = \begin{pmatrix} XYZ\_to\_LMX \end{pmatrix} \begin{pmatrix} X_2 \\ Y_2 \\ Z_2 \end{pmatrix} = \begin{pmatrix} 0.15514 & 0.54312 & -0.03286 \\ -0.15514 & 0.45684 & 0.03286 \\ 0 & 0 & 0.01608 \end{pmatrix} \begin{pmatrix} X_2 \\ Y_2 \\ Z_2 \end{pmatrix}.$$

The RGB to LMS matrix is the product of two matrices:

$$\begin{pmatrix} L \\ M \\ S \end{pmatrix} = (XYZ\_to\_LMS) (RGB\_to\_XYZ) \begin{pmatrix} R_2 \\ G_2 \\ B_2 \end{pmatrix} = (RGB\_to\_LMS) \begin{pmatrix} R_2 \\ G_2 \\ B_2 \end{pmatrix} = \begin{pmatrix} 17.8824 & 43.5161 & 4.11935 \\ 3.45565 & 27.1554 & 3.86714 \\ 0.0299566 & 0.184309 & 1.46709 \end{pmatrix} \begin{pmatrix} R_2 \\ G_2 \\ B_2 \end{pmatrix}. \quad (4)$$

4. The reduced colour domain is the plane including the origin, the white point, and the blue primary. Solving the plane equation for the origin ( $0, 0, 0$ ), the blue primary stimulus ( $L_B, M_B, S_B$ ) and the white stimulus ( $L_w, M_w, S_w$ ) gives the equation of the reduced plane

$$\alpha L + \beta M + \gamma S = 0$$

with

$$\alpha = M_w S_B - M_B S_w$$

TABLE II. DAC values of replacement colourmaps, which allow a designer to check the readability of colour images by protanopes and deuteranopes. As the plane of reduced stimuli is a diagonal plane of the RGB colour space of the video display, DAC values for red and green primaries are equal.

| Standard DAC values |     |     | Protan replacement values |                |                | Deutan replacement values |                |                | Standard DAC values |     |     | Protan replacement values |                |                | Deutan replacement values |                |                | Standard DAC values |     |     | Protan replacement values |                |                | Deutan replacement values |                |                |
|---------------------|-----|-----|---------------------------|----------------|----------------|---------------------------|----------------|----------------|---------------------|-----|-----|---------------------------|----------------|----------------|---------------------------|----------------|----------------|---------------------|-----|-----|---------------------------|----------------|----------------|---------------------------|----------------|----------------|
| I                   | J   | K   | I <sub>p</sub>            | J <sub>p</sub> | K <sub>p</sub> | I <sub>d</sub>            | J <sub>d</sub> | K <sub>d</sub> | I                   | J   | K   | I <sub>p</sub>            | J <sub>p</sub> | K <sub>p</sub> | I <sub>d</sub>            | J <sub>d</sub> | K <sub>d</sub> | I                   | J   | K   | I <sub>p</sub>            | J <sub>p</sub> | K <sub>p</sub> | I <sub>d</sub>            | J <sub>d</sub> | K <sub>d</sub> |
| 255                 | 255 | 255 | 255                       | 255            | 255            | 255                       | 255            | 255            | 255                 | 255 | 255 | 255                       | 255            | 255            | 255                       | 255            | 255            | 255                 | 255 | 255 | 255                       | 255            | 255            | 255                       | 255            | 255            |
| 204                 | 255 | 255 | 249                       | 254            | 254            | 239                       | 253            | 253            | 204                 | 255 | 253 | 249                       | 253            | 253            | 239                       | 253            | 253            | 204                 | 255 | 253 | 249                       | 253            | 253            | 239                       | 253            | 253            |
| 153                 | 255 | 255 | 246                       | 254            | 254            | 229                       | 254            | 254            | 153                 | 255 | 253 | 246                       | 253            | 253            | 229                       | 254            | 254            | 153                 | 255 | 253 | 246                       | 253            | 253            | 229                       | 254            | 254            |
| 102                 | 255 | 255 | 243                       | 254            | 254            | 222                       | 255            | 255            | 102                 | 255 | 253 | 243                       | 253            | 253            | 222                       | 255            | 255            | 102                 | 255 | 253 | 243                       | 253            | 253            | 222                       | 255            | 255            |
| 51                  | 255 | 255 | 242                       | 254            | 254            | 218                       | 255            | 255            | 51                  | 255 | 253 | 242                       | 252            | 252            | 218                       | 255            | 255            | 51                  | 255 | 253 | 242                       | 251            | 251            | 218                       | 255            | 255            |
| 0                   | 255 | 255 | 241                       | 254            | 254            | 217                       | 255            | 255            | 0                   | 255 | 253 | 241                       | 252            | 252            | 217                       | 255            | 255            | 0                   | 255 | 253 | 241                       | 251            | 251            | 217                       | 255            | 255            |
| 255                 | 204 | 255 | 210                       | 255            | 210            | 219                       | 252            | 253            | 255                 | 204 | 253 | 210                       | 254            | 254            | 219                       | 253            | 253            | 255                 | 204 | 253 | 210                       | 254            | 254            | 219                       | 253            | 253            |
| 204                 | 204 | 255 | 204                       | 255            | 204            | 203                       | 253            | 253            | 204                 | 204 | 253 | 204                       | 253            | 204            | 203                       | 253            | 253            | 204                 | 204 | 253 | 204                       | 253            | 204            | 203                       | 253            | 253            |
| 153                 | 204 | 255 | 199                       | 254            | 254            | 191                       | 253            | 253            | 153                 | 204 | 253 | 199                       | 253            | 253            | 191                       | 253            | 253            | 153                 | 204 | 253 | 199                       | 253            | 253            | 191                       | 253            | 253            |
| 102                 | 204 | 255 | 196                       | 254            | 254            | 181                       | 254            | 254            | 102                 | 204 | 253 | 196                       | 253            | 253            | 181                       | 254            | 254            | 102                 | 204 | 253 | 196                       | 252            | 252            | 181                       | 254            | 254            |
| 51                  | 204 | 255 | 194                       | 254            | 254            | 176                       | 254            | 254            | 51                  | 204 | 253 | 194                       | 253            | 253            | 176                       | 254            | 254            | 51                  | 204 | 253 | 194                       | 252            | 252            | 176                       | 254            | 254            |
| 0                   | 204 | 255 | 193                       | 254            | 254            | 175                       | 254            | 254            | 0                   | 204 | 253 | 193                       | 253            | 253            | 175                       | 254            | 254            | 0                   | 204 | 253 | 193                       | 252            | 252            | 175                       | 254            | 254            |
| 255                 | 153 | 255 | 168                       | 255            | 255            | 190                       | 251            | 251            | 255                 | 153 | 253 | 168                       | 253            | 253            | 190                       | 252            | 253            | 255                 | 153 | 253 | 168                       | 253            | 253            | 190                       | 252            | 253            |
| 204                 | 153 | 255 | 160                       | 255            | 255            | 171                       | 252            | 252            | 204                 | 153 | 253 | 160                       | 253            | 253            | 171                       | 253            | 253            | 204                 | 153 | 253 | 160                       | 253            | 253            | 171                       | 253            | 253            |
| 153                 | 153 | 255 | 153                       | 255            | 255            | 155                       | 253            | 253            | 153                 | 153 | 253 | 153                       | 253            | 253            | 155                       | 253            | 253            | 153                 | 153 | 253 | 153                       | 253            | 253            | 155                       | 253            | 253            |
| 102                 | 153 | 255 | 149                       | 254            | 254            | 143                       | 253            | 253            | 102                 | 153 | 253 | 149                       | 253            | 253            | 143                       | 253            | 253            | 102                 | 153 | 253 | 149                       | 253            | 253            | 143                       | 253            | 253            |
| 51                  | 153 | 255 | 146                       | 254            | 254            | 136                       | 253            | 253            | 51                  | 153 | 253 | 146                       | 253            | 253            | 136                       | 253            | 253            | 51                  | 153 | 253 | 146                       | 253            | 253            | 136                       | 253            | 253            |
| 0                   | 153 | 255 | 145                       | 254            | 254            | 134                       | 253            | 253            | 0                   | 153 | 253 | 145                       | 253            | 253            | 134                       | 253            | 253            | 0                   | 153 | 253 | 145                       | 253            | 253            | 134                       | 253            | 253            |
| 255                 | 102 | 255 | 131                       | 255            | 255            | 167                       | 250            | 250            | 255                 | 102 | 253 | 131                       | 254            | 254            | 167                       | 251            | 251            | 255                 | 102 | 253 | 131                       | 254            | 254            | 167                       | 251            | 251            |
| 204                 | 102 | 255 | 120                       | 255            | 255            | 144                       | 251            | 251            | 204                 | 102 | 253 | 120                       | 253            | 253            | 144                       | 252            | 252            | 204                 | 102 | 253 | 120                       | 253            | 253            | 144                       | 252            | 252            |
| 153                 | 102 | 255 | 110                       | 255            | 255            | 123                       | 252            | 252            | 153                 | 102 | 253 | 110                       | 253            | 253            | 123                       | 253            | 253            | 153                 | 102 | 253 | 110                       | 253            | 253            | 123                       | 253            | 253            |
| 102                 | 102 | 255 | 103                       | 255            | 255            | 107                       | 253            | 253            | 102                 | 102 | 253 | 103                       | 253            | 253            | 107                       | 253            | 253            | 102                 | 102 | 253 | 103                       | 253            | 253            | 107                       | 253            | 253            |
| 51                  | 102 | 255 | 99                        | 254            | 254            | 97                        | 253            | 253            | 51                  | 102 | 253 | 99                        | 253            | 253            | 97                        | 253            | 253            | 51                  | 102 | 253 | 99                        | 253            | 253            | 97                        | 253            | 253            |
| 0                   | 102 | 255 | 98                        | 254            | 254            | 94                        | 253            | 253            | 0                   | 102 | 253 | 98                        | 253            | 253            | 94                        | 253            | 253            | 0                   | 102 | 253 | 98                        | 253            | 253            | 94                        | 253            | 253            |
| 255                 | 51  | 255 | 105                       | 255            | 255            | 152                       | 250            | 250            | 255                 | 51  | 253 | 105                       | 253            | 253            | 152                       | 250            | 250            | 255                 | 51  | 253 | 105                       | 253            | 253            | 152                       | 250            | 250            |
| 204                 | 51  | 255 | 89                        | 255            | 255            | 126                       | 251            | 251            | 204                 | 51  | 253 | 89                        | 253            | 253            | 126                       | 252            | 252            | 204                 | 51  | 253 | 89                        | 253            | 253            | 126                       | 252            | 252            |
| 153                 | 51  | 255 | 74                        | 255            | 255            | 102                       | 252            | 252            | 153                 | 51  | 253 | 74                        | 253            | 253            | 102                       | 253            | 253            | 153                 | 51  | 253 | 74                        | 253            | 253            | 102                       | 252            | 252            |
| 102                 | 51  | 255 | 62                        | 255            | 255            | 80                        | 252            | 252            | 102                 | 51  | 253 | 62                        | 253            | 253            | 80                        | 253            | 253            | 102                 | 51  | 253 | 62                        | 253            | 253            | 80                        | 252            | 252            |
| 51                  | 51  | 255 | 54                        | 255            | 255            | 65                        | 253            | 253            | 51                  | 51  | 253 | 54                        | 253            | 253            | 65                        | 253            | 253            | 51                  | 51  | 253 | 54                        | 253            | 253            | 65                        | 253            | 253            |
| 0                   | 51  | 255 | 51                        | 255            | 255            | 60                        | 253            | 253            | 0                   | 51  | 253 | 51                        | 253            | 253            | 60                        | 253            | 253            | 0                   | 51  | 253 | 51                        | 253            | 253            | 60                        | 253            | 253            |
| 255                 | 0   | 255 | 96                        | 255            | 255            | 148                       | 250            | 250            | 255                 | 0   | 253 | 96                        | 253            | 253            | 148                       | 250            | 250            | 255                 | 0   | 253 | 96                        | 253            | 253            | 148                       | 250            | 250            |
| 204                 | 0   | 255 | 77                        | 255            | 255            | 121                       | 251            | 251            | 204                 | 0   | 253 | 77                        | 253            | 253            | 121                       | 251            | 251            | 204                 | 0   | 253 | 77                        | 253            | 253            | 121                       | 251            | 251            |
| 153                 | 0   | 255 | 59                        | 255            | 255            | 94                        | 252            | 252            | 153                 | 0   | 253 | 59                        | 253            | 253            | 94                        | 252            | 252            | 153                 | 0   | 253 | 59                        | 253            | 253            | 94                        | 252            | 252            |
| 102                 | 0   | 255 | 42                        | 255            | 255            | 70                        | 252            | 252            | 102                 | 0   | 253 | 42                        | 253            | 253            | 70                        | 252            | 252            | 102                 | 0   | 253 | 42                        | 253            | 253            | 70                        | 252            | 252            |
| 51                  | 0   | 255 | 27                        | 255            | 255            | 51                        | 252            | 252            | 51                  | 0   | 253 | 27                        | 253            | 253            | 51                        | 252            | 252            | 51                  | 0   | 253 | 27                        | 253            | 253            | 51                        | 252            | 252            |
| 0                   | 0   | 255 | 21                        | 255            | 255            | 44                        | 253            | 253            | 0                   | 0   | 253 | 21                        | 253            | 253            | 44                        | 253            | 253            | 0                   | 0   | 253 | 21                        | 253            | 253            | 44                        | 253            | 253            |

TABLE II. (Continued)

| Standard DAC values |     |     | Protan replacement values |       |            | Deutan replacement values |     |     | Standard DAC values |     |     | Protan replacement values |            |       | Deutan replacement values |       |     | Standard DAC values |     |     | Protan replacement values |     |            | Deutan replacement values |            |       |
|---------------------|-----|-----|---------------------------|-------|------------|---------------------------|-----|-----|---------------------|-----|-----|---------------------------|------------|-------|---------------------------|-------|-----|---------------------|-----|-----|---------------------------|-----|------------|---------------------------|------------|-------|
| $I$                 | $J$ | $K$ | $I_p, J_p$                | $K_p$ | $I_d, J_d$ | $K_d$                     | $I$ | $J$ | $K$                 | $I$ | $J$ | $K$                       | $I_p, J_p$ | $K_p$ | $I_d, J_d$                | $K_d$ | $I$ | $J$                 | $K$ | $I$ | $J$                       | $K$ | $I_p, J_p$ | $K_p$                     | $I_d, J_d$ | $K_d$ |
| 255                 | 255 | 204 | 255                       | 204   | 253        | 203                       | 255 | 255 | 102                 | 255 | 255 | 102                       | 255        | 103   | 253                       | 107   | 255 | 255                 | 0   | 255 | 255                       | 0   | 255        | 21                        | 253        | 44    |
| 204                 | 255 | 204 | 249                       | 204   | 239        | 205                       | 204 | 255 | 102                 | 249 | 255 | 102                       | 249        | 102   | 239                       | 110   | 204 | 255                 | 0   | 249 | 255                       | 0   | 249        | 17                        | 239        | 52    |
| 153                 | 255 | 204 | 246                       | 203   | 229        | 205                       | 153 | 255 | 102                 | 246 | 255 | 102                       | 246        | 102   | 229                       | 112   | 153 | 255                 | 0   | 246 | 255                       | 0   | 246        | 12                        | 229        | 56    |
| 102                 | 255 | 204 | 243                       | 203   | 222        | 206                       | 102 | 255 | 102                 | 243 | 255 | 102                       | 243        | 102   | 222                       | 113   | 102 | 255                 | 0   | 243 | 255                       | 0   | 243        | 8                         | 222        | 59    |
| 51                  | 255 | 204 | 242                       | 203   | 218        | 206                       | 51  | 255 | 102                 | 242 | 255 | 102                       | 242        | 102   | 218                       | 114   | 51  | 255                 | 0   | 242 | 255                       | 0   | 242        | 4                         | 218        | 60    |
| 0                   | 255 | 204 | 241                       | 203   | 217        | 206                       | 0   | 255 | 102                 | 241 | 255 | 102                       | 241        | 102   | 217                       | 114   | 0   | 255                 | 0   | 241 | 255                       | 0   | 241        | 0                         | 217        | 61    |
| 255                 | 204 | 204 | 210                       | 204   | 219        | 202                       | 255 | 204 | 102                 | 210 | 204 | 102                       | 210        | 104   | 219                       | 105   | 255 | 204                 | 0   | 210 | 204                       | 0   | 210        | 24                        | 219        | 36    |
| 204                 | 204 | 204 | 204                       | 204   | 203        | 203                       | 204 | 204 | 102                 | 204 | 204 | 102                       | 204        | 103   | 203                       | 107   | 204 | 204                 | 0   | 204 | 204                       | 0   | 204        | 21                        | 203        | 44    |
| 153                 | 204 | 204 | 199                       | 204   | 191        | 204                       | 153 | 204 | 102                 | 199 | 204 | 102                       | 199        | 103   | 191                       | 109   | 153 | 204                 | 0   | 199 | 204                       | 0   | 199        | 18                        | 191        | 50    |
| 102                 | 204 | 204 | 196                       | 204   | 181        | 205                       | 102 | 204 | 102                 | 196 | 204 | 102                       | 196        | 102   | 181                       | 111   | 102 | 204                 | 0   | 196 | 204                       | 0   | 196        | 15                        | 181        | 53    |
| 51                  | 204 | 204 | 194                       | 204   | 176        | 205                       | 51  | 204 | 102                 | 194 | 204 | 102                       | 194        | 102   | 176                       | 111   | 51  | 204                 | 0   | 194 | 204                       | 0   | 194        | 14                        | 176        | 55    |
| 0                   | 204 | 204 | 193                       | 203   | 175        | 205                       | 0   | 204 | 102                 | 193 | 204 | 102                       | 193        | 102   | 175                       | 111   | 0   | 204                 | 0   | 193 | 204                       | 0   | 193        | 13                        | 175        | 55    |
| 255                 | 153 | 204 | 168                       | 204   | 190        | 201                       | 255 | 153 | 102                 | 168 | 204 | 102                       | 168        | 104   | 190                       | 102   | 255 | 153                 | 0   | 168 | 204                       | 0   | 168        | 26                        | 190        | 27    |
| 204                 | 153 | 204 | 160                       | 204   | 171        | 202                       | 204 | 153 | 102                 | 160 | 204 | 102                       | 160        | 103   | 171                       | 105   | 204 | 153                 | 0   | 160 | 204                       | 0   | 160        | 23                        | 171        | 38    |
| 153                 | 153 | 204 | 153                       | 204   | 155        | 203                       | 153 | 153 | 102                 | 153 | 153 | 102                       | 153        | 103   | 155                       | 107   | 153 | 153                 | 0   | 153 | 204                       | 0   | 153        | 21                        | 155        | 44    |
| 102                 | 153 | 204 | 149                       | 204   | 143        | 204                       | 102 | 153 | 102                 | 149 | 204 | 102                       | 149        | 103   | 143                       | 109   | 102 | 153                 | 0   | 149 | 204                       | 0   | 149        | 19                        | 143        | 48    |
| 51                  | 153 | 204 | 146                       | 204   | 136        | 204                       | 51  | 153 | 102                 | 146 | 204 | 102                       | 146        | 103   | 136                       | 109   | 51  | 153                 | 0   | 146 | 204                       | 0   | 146        | 18                        | 136        | 50    |
| 0                   | 153 | 204 | 145                       | 204   | 134        | 204                       | 0   | 153 | 102                 | 145 | 204 | 102                       | 145        | 103   | 134                       | 110   | 0   | 153                 | 0   | 145 | 204                       | 0   | 145        | 17                        | 134        | 50    |
| 255                 | 102 | 204 | 131                       | 204   | 167        | 200                       | 255 | 102 | 102                 | 131 | 204 | 102                       | 131        | 104   | 167                       | 101   | 255 | 102                 | 0   | 131 | 204                       | 0   | 131        | 27                        | 167        | 18    |
| 204                 | 102 | 204 | 120                       | 204   | 144        | 202                       | 204 | 102 | 102                 | 120 | 204 | 102                       | 120        | 104   | 144                       | 104   | 204 | 102                 | 0   | 120 | 204                       | 0   | 120        | 25                        | 144        | 33    |
| 153                 | 102 | 204 | 110                       | 204   | 123        | 203                       | 153 | 102 | 102                 | 110 | 204 | 102                       | 110        | 103   | 123                       | 106   | 153 | 102                 | 0   | 110 | 204                       | 0   | 110        | 22                        | 123        | 40    |
| 102                 | 102 | 204 | 103                       | 204   | 107        | 203                       | 102 | 102 | 102                 | 103 | 204 | 102                       | 103        | 103   | 107                       | 107   | 102 | 102                 | 0   | 103 | 204                       | 0   | 103        | 21                        | 107        | 44    |
| 51                  | 102 | 204 | 99                        | 204   | 97         | 204                       | 51  | 102 | 102                 | 99  | 204 | 102                       | 99         | 103   | 97                        | 108   | 51  | 102                 | 0   | 99  | 204                       | 0   | 99         | 20                        | 97         | 46    |
| 0                   | 102 | 204 | 98                        | 204   | 94         | 204                       | 0   | 102 | 102                 | 98  | 204 | 102                       | 98         | 103   | 94                        | 108   | 0   | 102                 | 0   | 98  | 204                       | 0   | 98         | 19                        | 94         | 47    |
| 255                 | 51  | 204 | 105                       | 204   | 152        | 200                       | 255 | 51  | 102                 | 105 | 204 | 102                       | 105        | 104   | 152                       | 100   | 255 | 51                  | 0   | 105 | 204                       | 0   | 105        | 28                        | 152        | 9     |
| 204                 | 51  | 204 | 89                        | 204   | 126        | 201                       | 204 | 51  | 102                 | 89  | 204 | 102                       | 89         | 104   | 126                       | 103   | 204 | 51                  | 0   | 89  | 204                       | 0   | 89         | 25                        | 126        | 30    |
| 153                 | 51  | 204 | 74                        | 204   | 102        | 202                       | 153 | 51  | 102                 | 74  | 204 | 102                       | 74         | 103   | 102                       | 105   | 153 | 51                  | 0   | 74  | 204                       | 0   | 74         | 23                        | 102        | 38    |
| 102                 | 51  | 204 | 62                        | 204   | 80         | 203                       | 102 | 51  | 102                 | 62  | 204 | 102                       | 62         | 103   | 80                        | 107   | 102 | 51                  | 0   | 62  | 204                       | 0   | 62         | 22                        | 80         | 42    |
| 51                  | 51  | 204 | 54                        | 204   | 65         | 203                       | 51  | 51  | 102                 | 54  | 204 | 102                       | 54         | 103   | 65                        | 107   | 51  | 51                  | 0   | 54  | 204                       | 0   | 54         | 21                        | 65         | 44    |
| 0                   | 51  | 204 | 51                        | 204   | 60         | 203                       | 0   | 51  | 102                 | 51  | 204 | 102                       | 51         | 103   | 60                        | 107   | 0   | 51                  | 0   | 51  | 204                       | 0   | 51         | 20                        | 60         | 45    |
| 255                 | 0   | 204 | 96                        | 204   | 148        | 200                       | 255 | 0   | 102                 | 96  | 204 | 102                       | 96         | 104   | 148                       | 100   | 255 | 0                   | 0   | 96  | 204                       | 0   | 96         | 28                        | 148        | 0     |
| 204                 | 0   | 204 | 77                        | 204   | 121        | 201                       | 204 | 0   | 102                 | 77  | 204 | 102                       | 77         | 104   | 121                       | 103   | 204 | 0                   | 0   | 77  | 204                       | 0   | 77         | 26                        | 121        | 29    |
| 153                 | 0   | 204 | 59                        | 204   | 94         | 202                       | 153 | 0   | 102                 | 59  | 204 | 102                       | 59         | 103   | 94                        | 105   | 153 | 0                   | 0   | 59  | 204                       | 0   | 59         | 23                        | 94         | 37    |
| 102                 | 0   | 204 | 42                        | 204   | 70         | 203                       | 102 | 0   | 102                 | 42  | 204 | 102                       | 42         | 103   | 70                        | 106   | 102 | 0                   | 0   | 42  | 204                       | 0   | 42         | 22                        | 70         | 42    |
| 51                  | 0   | 204 | 27                        | 204   | 51         | 203                       | 51  | 0   | 102                 | 27  | 204 | 102                       | 27         | 103   | 51                        | 107   | 51  | 0                   | 0   | 27  | 204                       | 0   | 27         | 21                        | 51         | 44    |
| 0                   | 0   | 204 | 21                        | 204   | 44         | 203                       | 0   | 0   | 102                 | 21  | 204 | 102                       | 21         | 103   | 44                        | 107   | 0   | 0                   | 0   | 21  | 204                       | 0   | 21         | 21                        | 44         | 44    |

$$\beta = S_W L_B - S_B L_W$$

$$\gamma = L_W M_B - L_B M_W.$$

The reduction of the normal colour domain to the dichromatic colour domain maintains the fundamental values corresponding to the existing photopigments,  $S$  and  $M$  for the protanope, and  $S$  and  $L$  for the deuteranope.

The replacement tristimulus value corresponding to the missing photopigment is, for the protanope,

$$L_p = -(\beta M + \gamma S)/\alpha;$$

and, for the deuteranope,

$$M_d = -(\alpha L + \gamma S)/\beta.$$

This results in the following linear transformations for reducing the normal colour domain to the dichromat colour domain, for protanopes:

$$\begin{pmatrix} L_p \\ M_p \\ S_p \end{pmatrix} = \begin{pmatrix} 0 & 2.02344 & -2.52581 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} L \\ M \\ S \end{pmatrix} \quad (5)$$

and for deuteranopes:

$$\begin{pmatrix} L_d \\ M_d \\ S_d \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0.494207 & 0 & 1.24827 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} L \\ M \\ S \end{pmatrix}. \quad (5)$$

5. Transformation of  $L_d M_d S_d$  or  $L_p M_p S_p$  to RGB is obtained using the inverse matrix of matrix (4) in step 3):

$$\begin{pmatrix} R_d \\ G_d \\ B_d \end{pmatrix} = (RGB\_to\_LMS)^{-1} \begin{pmatrix} L_d \\ M_d \\ S_d \end{pmatrix}$$

$$= \begin{pmatrix} 0.080944 & -0.130504 & 0.116721 \\ -0.0102485 & 0.0540194 & -0.113615 \\ -0.000365294 & -0.00412163 & 0.693513 \end{pmatrix} \times \begin{pmatrix} L_d \\ M_d \\ S_d \end{pmatrix}. \quad (6)$$

6. DAC-values of the replacement colourmaps are obtained using the inverse of the relationship described in step (1):

$$I_d = 255 R_d^{(1/2.2)} \quad (7)$$

$$J_d = 255 G_d^{(1/2.2)} \quad (7)$$

$$K_d = 255 B_d^{(1/2.2)}. \quad (7)$$

Table II shows the DAC values of the resulting replacement colourmaps.

Because the plane of reduced stimuli is a diagonal plane of the RGB colour space of the video display, DAC values for red and green primaries are equal.

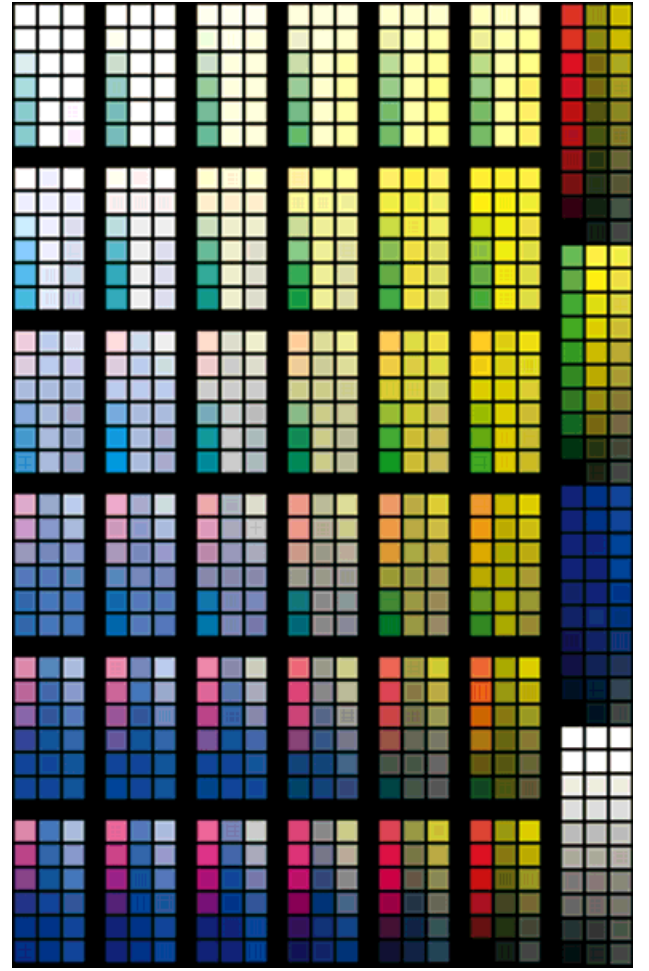


FIG. 3. Colour illustration of the 256 colour palettes specified in Table II. Within each block, the first column illustrates the original colour of the normal colourmap, the second and third columns illustrate its appearance in the protan and deutan transformation. The vertical sequence of samples is as in Table II.

Figure 3 illustrates side-by-side every sample of the 256 colour palette of Table II in its normal, protan, and deutan version, in the same order as in the table. It can be seen that samples of the colourful left column are replaced by yellow or blue shades, in various lightnesses or saturations. It can be seen also that the protan and the deutan transformation yield colours of different lightnesses.

Figure 4 shows an example where the normal, protan, and deutan versions of an RATP-RER transportation map of Paris (France) are presented in colour. The Réseau Express Régional (RER) of Régie Autonome des Transports Parisiens (RATP, Paris, France) is the underground and train network, which operates between Paris and its outskirts. All colours of the transportation map that have been originally used by the designer are part of Table II. The protan and deutan versions result from replacing the normal colourmap by the protan or by the deutan colourmap. Although our algorithm was not available by the time the map was designed, the various transportation



FIG. 4. Normal, protan, and deutan versions of the transportation map of the RATP-RER (Paris, France). The original image has been produced using a high-resolution version of the map provided by RATP (by courtesy of RATP). Every colour has been adjusted in order to comply with the Web version accessible at: <http://www.ratp.fr/Transpor/Reseaux/planrer.htm>. All colours are part of Table II. The protan and deutan versions result from the replacement of the normal colourmap by the protan or by the deutan colourmap. The original red transportation line is transformed in a darker shade in the protan version than in the deutan version. Left: normal version. Upper-right: protan version. Lower-right: deutan version.

lines are still identifiable. Now that it is available, guidance on the choice of colours can be obtained simply by substituting the colourmaps.

## DISCUSSION

Although the colourmaps we propose have been constructed along the same lines as those used in the full “TrueColour” representation,<sup>6</sup> we should evaluate how far the simulation of dichromatic vision is affected by the simplification of the algorithm that we have adopted here.

## Colorimetric Accuracy

When proper calibration of the user’s video display is not available, the lack of conformity with the ITU standard may be a source of discrepancy. We have computed the errors of confused colours originating from poor calibration taking as a model the most common shifts of CRT parameters. In Table III, we indicate the errors in terms of DAC values for a few samples of the protan replacement palette:

if the primaries and reference white have the chromaticity recommended by the National Television Systems Committee<sup>12</sup> instead of those recommended by ITU (Table I),

TABLE III. Computed DAC values,  $I_p J_p K_p$  for the protan replacement colourmap, considering different CRT primaries, reference white, and transfer functions. First column: I J K refer to a few DAC values selected from the original colourmap. Following columns:  $I_p J_p K_p$  are obtained by performing steps 1–7 of the computation as indicated in the block diagram in Fig. 2. Last line gives the scaling factor used to compute the replacement DAC values in each case, which at the same time is the maximum achievable scaling factor enabling the reduction of the entire colour gamut within the real colour gamut of the monitor.

| DAC values of the original colourmap |     |     | DAC values for the protan replacement map |       |                                           |        |                                                     |         |                                                     |          |
|--------------------------------------|-----|-----|-------------------------------------------|-------|-------------------------------------------|--------|-----------------------------------------------------|---------|-----------------------------------------------------|----------|
|                                      |     |     | ITU standard<br>D65 white<br>Gamma 2.2    |       | NTSC standard<br>CIE C white<br>Gamma 2.2 |        | ITU-R BT.709<br>primaries<br>D93 white<br>Gamma 2.2 |         | ITU-R BT.709<br>primaries<br>D65 white<br>Gamma 1.8 |          |
| I                                    | J   | K   | $I_p, J_p$                                | $K_p$ | $I'_p, J'_p$                              | $K'_p$ | $I''_p, J''_p$                                      | $K''_p$ | $I'''_p, J'''_p$                                    | $K'''_p$ |
| 255                                  | 255 | 255 | 255                                       | 255   | 254                                       | 254    | 255                                                 | 255     | 254                                                 | 254      |
| 0                                    | 255 | 255 | 241                                       | 254   | 235                                       | 255    | 243                                                 | 254     | 238                                                 | 254      |
| 255                                  | 0   | 255 | 96                                        | 255   | 112                                       | 253    | 89                                                  | 255     | 77                                                  | 255      |
| 0                                    | 0   | 255 | 21                                        | 255   | 30                                        | 254    | 17                                                  | 255     | 12                                                  | 254      |
| 255                                  | 255 | 0   | 255                                       | 21    | 254                                       | 30     | 255                                                 | 17      | 254                                                 | 12       |
| 0                                    | 255 | 0   | 241                                       | 0     | 235                                       | 41     | 243                                                 | 0       | 238                                                 | 0        |
| 255                                  | 0   | 0   | 96                                        | 28    | 112                                       | 0      | 89                                                  | 23      | 77                                                  | 17       |
| 0                                    | 0   | 0   | 21                                        | 21    | 30                                        | 30     | 17                                                  | 17      | 12                                                  | 12       |
| 170                                  | 0   | 0   | 65                                        | 24    | 77                                        | 24     | 60                                                  | 20      | 52                                                  | 15       |
| 85                                   | 0   | 0   | 37                                        | 21    | 46                                        | 29     | 33                                                  | 18      | 29                                                  | 13       |
| 0                                    | 170 | 0   | 161                                       | 16    | 158                                       | 35     | 163                                                 | 13      | 159                                                 | 8        |
| 0                                    | 85  | 0   | 82                                        | 20    | 82                                        | 31     | 82                                                  | 16      | 81                                                  | 11       |
| 0                                    | 0   | 170 | 21                                        | 170   | 30                                        | 170    | 17                                                  | 170     | 12                                                  | 170      |
| 0                                    | 0   | 85  | 21                                        | 86    | 30                                        | 88     | 17                                                  | 86      | 12                                                  | 86       |
| Scaling factor                       |     |     | 0.992052                                  |       | 0.982004                                  |        | 0.994881                                            |         | 0.992052                                            |          |

if D93 reference illuminant<sup>13</sup> is used as the nominal white instead of D65,  
if the gamma value is 1.8 instead of 2.2.

This gives a figure of merit of the replacement colourmaps. It shows that significant discrepancies could arise from changing either the primaries or the video transfer function; the former should not occur, since all ITU members have agreed on a unique recommendation that is gaining acceptance for CRT based applications,<sup>8</sup> but the latter is easily encountered in practice. However, changing the nominal white would have minimal effect on the palette.

### Simplification of the Reduction Scheme

We propose replacement colourmaps for simulating protanopic vision and deuteranopic vision, which are the most severe cases of colour deficiency. Anomalous trichromatic observers do not confuse all colours of a colour confusion line, so they would be able to discriminate the colours that

are indiscriminable by a dichromatic observer of the same type.

In the absence of an officially recommended set of cone fundamentals, either those of Smith and Pokorny<sup>9</sup> or those of Stockman, MacLeod, and Johnson<sup>14</sup> are currently employed in research. We assume that both come close to the average normal observer.

Strictly, the Smith and Pokorny transformation applies to modified tristimulus values obtained from the spectral power distribution of the stimulus and the colour-matching functions modified by Judd and Vos.<sup>11</sup> Because the ITU video display standard does not specify the primaries in terms of spectral power distribution but in terms of CIE 1931 chromaticity coordinates, we have extended the application of the Vos formula to the chromaticity coordinates of the primaries and nominal white. In order to evaluate the error introduced by this procedure, we have compared the results with those obtained by a rigorous calculation for an actual video display.

First, we have measured the  $(x, y)$  chromaticity coordi-

TABLE IV. Comparison between measurements and calculations for an actual CRT video display (IIYAMA MF-8617A 1995). Columns 2–4:  $(x, y)$  chromaticity coordinates and  $(Y)$  luminance (normalized to 100) in the CIE 1931 colorimetric system. Column 5–7:  $(x', y')$  chromaticity coordinates and  $(Y_m)$  luminance obtained using the Judd–Vos modified colour-matching functions. Column 8–9:  $(x', y')$  chromaticity coordinates obtained applying the Vos formula to  $(x, y)$  CIE 1931 chromaticity coordinates.

|       | Measurements |        |         | Results obtained using Vos cmfs |        |         | $x' y'$ converted from $x y$ |        |
|-------|--------------|--------|---------|---------------------------------|--------|---------|------------------------------|--------|
|       | $x$          | $y$    | $Y$     | $x'$                            | $y'$   | $Y_m$   | $x'$                         | $y'$   |
| RED   | 0.6254       | 0.3370 | 24.4929 | 0.6242                          | 0.3406 | 24.5046 | 0.6241                       | 0.3396 |
| GREEN | 0.2818       | 0.6006 | 67.8719 | 0.2838                          | 0.6052 | 67.8903 | 0.2837                       | 0.6017 |
| BLUE  | 0.1500       | 0.0646 | 7.6353  | 0.1545                          | 0.0727 | 8.1588  | 0.1530                       | 0.0727 |
| WHITE | 0.3127       | 0.3290 | 100     | 0.3175                          | 0.3394 | 100.55  | 0.3157                       | 0.3345 |

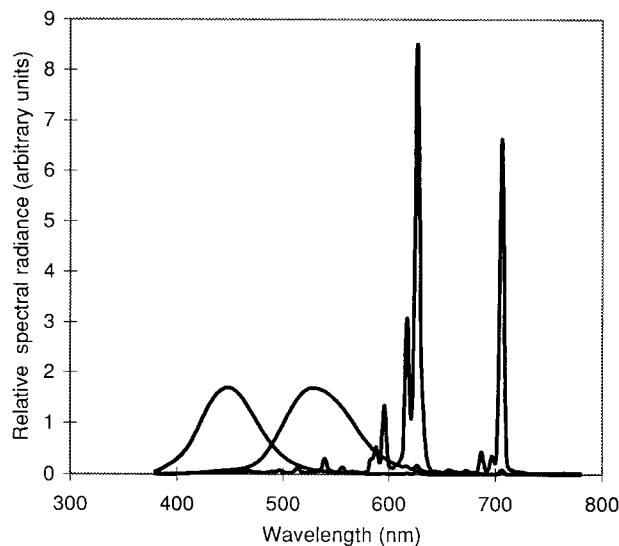


FIG. 5. Relative spectral power distribution of the primaries of a CRT video display (IIYAMA MF-8617A 1995) adjusted to give a metamer of D65 as the nominal white.

nates of four CRT video displays and selected the one (IIYAMA MF-8617A 1995) that best conformed to the ITU standard (Table IV). Then we have measured the spectral power distribution of its primaries every 5 nm using a calibrated telespectrophotometer (Minolta CS-1000), and we have computed the absolute spectral power distribution (Fig. 5) and the ( $Y$ ) luminance of the primaries that gives a metamer of D65 as the nominal white (Table IV). Knowing the absolute spectral power distribution of the primaries, it is possible to calculate the ( $X'$ ,  $Y'$ ,  $Z'$ ) modified tristimulus values and the ( $x'$ ,  $y'$ ) modified chromaticity coordinates of the primaries and the white for the Judd–Vos modified colorimetric observer, using the modified colour-matching functions given by Vos.<sup>11</sup> We have also computed the result of applying the Vos modification<sup>11</sup> (step 3 of the procedure

described in the methods section) to the CIE 1931 ( $x$ ,  $y$ ) chromaticity coordinates of the measured video display. It appears that the values differ only on the third digit. Then we have computed the full palette for this actual monitor using the procedure described in the Methods section, and the full palette for the same monitor starting the transformation with the ( $x'$ ,  $y'$ ) modified chromaticity coordinates directly obtained from the absolute spectral power distribution of the primaries and the white. For every element of the 256 colour palette, the difference in DAC-value is never larger than one (Table V). This leads us to conclude that, for our particular application, extending the Vos formula to the primaries and the white of a CRT video display is satisfactory.

We are aware that several sources of inter-observer variability<sup>15</sup> such as lens pigmentation, macular pigmentation, cone effective optical density, and spectral sensitivity of the visual photopigments are also ignored in this simplified scheme for illustrating the losses of dichromatic colour vision.

Finally, by adopting the diagonal plane on which to project the confused colours, we slightly distort the colour appearance of the simulated dichromatic image, compared to our previous simulation. In terms of dominant wavelengths, this corresponds to a shift from 475 to 464 nm for the blue anchor wavelength, and to a shift from 575 to 571 nm for the yellow anchor wavelength.

## CONCLUSION

In conclusion, replacing a normal palette by a reduced palette allows the designer to check the readability of colour information by dichromatic observers in any displayed image. Although an accurate simulation of dichromatic vision would require a careful calibration of the video display, our colourmaps provide an immediate and efficient warning of possible losses of readability by

TABLE V. DAC values,  $I_p J_p K_p$ , for the protan replacement colourmap computed for the actual CRT video display (IIYAMA MF-8617A 1995), obtained using the Judd–Vos modified colour-matching functions with the spectral distribution of the primaries, or obtained applying the Vos formula to ( $x$ ,  $y$ ) CIE 1931 chromaticity coordinates.

| DAC values of the original colourmap |     |     | DAC values obtained using Vos cmfs |       | DAC values obtained applying Vos formula |        |
|--------------------------------------|-----|-----|------------------------------------|-------|------------------------------------------|--------|
| $I$                                  | $J$ | $K$ | $I_p, J_p$                         | $K_p$ | $I'_p, J'_p$                             | $K'_p$ |
| 255                                  | 255 | 255 | 254                                | 254   | 254                                      | 254    |
| 0                                    | 255 | 255 | 238                                | 254   | 238                                      | 254    |
| 255                                  | 0   | 255 | 105                                | 255   | 106                                      | 255    |
| 0                                    | 0   | 255 | 23                                 | 254   | 23                                       | 254    |
| 255                                  | 255 | 0   | 254                                | 23    | 254                                      | 23     |
| 0                                    | 255 | 0   | 238                                | 0     | 238                                      | 0      |
| 255                                  | 0   | 0   | 105                                | 32    | 106                                      | 32     |
| 0                                    | 0   | 0   | 23                                 | 23    | 23                                       | 23     |
| 170                                  | 0   | 0   | 72                                 | 27    | 72                                       | 27     |
| 85                                   | 0   | 0   | 40                                 | 24    | 41                                       | 24     |
| 0                                    | 170 | 0   | 159                                | 18    | 159                                      | 18     |
| 0                                    | 85  | 0   | 81                                 | 22    | 81                                       | 22     |
| 0                                    | 0   | 170 | 23                                 | 170   | 23                                       | 170    |
| 0                                    | 0   | 85  | 23                                 | 87    | 23                                       | 87     |
| Scaling factor                       |     |     | 0.989729                           |       | 0.989725                                 |        |

colour-deficients. This tool could be used to check the legibility of transport colour signals, control panels, and information displays.

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We thank Mr C. Vuaillet of Réseau Autonome des Transports Parisiens (RATP) for providing the high-resolution digital file of the Réseau Express Régional (RER) transportation map. The data of Table II as well as example images and an interactive Java applet to compute further replacement values for any DAC values combination are accessible at <http://tsi.enst.fr/~brettel/colourmaps.html>.

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